2	**(a)**	use of kelvin temperatures	B1	
		both values of (V/T) correct (11.87), V/T is constant so pressure is constant	M1	[2]
		(allow use of n =1. Do not allow other values of n.)		
	(b)	**(i)** work done $= p\Delta V$		
		$= 4.2 \times 10^5 \times (3.87 - 3.49) \times 10^3 \times 10^{-6}$	C1	
		$= 160$ J	A1	[2]
		(do not allow use of V instead of ΔV)		
		(ii) increase/change in internal energy = heating of system		
		+ work done on system	C1	
		$= 565 - 160$		
		$= 405$ J	A1	[2]
	(c)	internal energy = sum of kinetic energy and potential energy / $E_K + E_P$	B1	
		no intermolecular forces	M1	
		no potential energy (so $\Delta U = \Delta E_K$)	A1	[3]